



RESPIRATORY SYSTEM ORGANS AND DISORDERS THAT AFFLICTS THE CIRCULATORY
AND RESPIRATORY SYSTEM

LEARNING MODULE

NAME: _____

YEAR AND SECTION : _____



LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. identify and describe the parts of the respiratory system.
2. explain the mechanism on how the respiratory and circulatory systems work together to transport nutrients, gases, and molecules to and from the different parts of the body.
3. infer how one's lifestyle can affect the functioning of the respiratory and circulatory systems

MOTIVATIONAL ACTIVITY. All four pictures in each given set depict negative ways of living. They are connected by one common word that indicates the effect of one's lifestyle on the functioning of the respiratory and circulatory systems. The expected answers are illnesses that are brought about by the negative lifestyles. Write your answers in the box provided for each number



WHAT'S THE WORD?

A _____



WHAT'S THE WORD?

H _____



LEARN MORE!

RESPIRATORY SYSTEM is the biological system of an organism that introduces respiratory gases to the interior and performs gas exchange.

BREATHING is the process of drawing air into the lungs and its corresponding expulsion. During inhalation, oxygen enters the lungs and during exhalation, carbon dioxide exits the lungs. The respiratory system includes many different organs that work together for gas exchange to take place.

NOSE is the gateway of the respiratory system. Air enters the body via the nose. The nasal cavity is lined with tiny hairs called cilia.

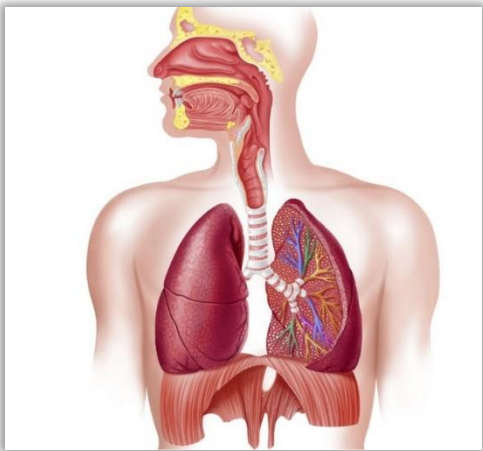


Figure 1. Respiratory System

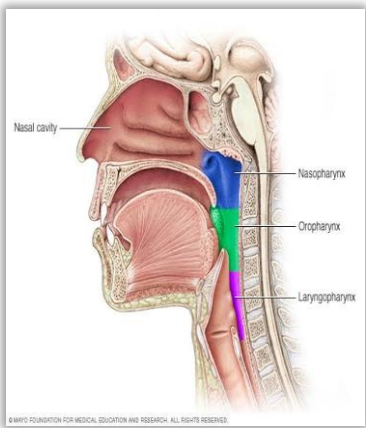


Figure 1.1. Pharynx

PHARYNX or **THROAT** is the narrow space located right after the mouth and nose. The part of the pharynx that is related to respiration is called *nasopharynx*.

LARYNX or **VOICE BOX** is the organ that connects the pharynx and the trachea. It contains the vocal chords that produce sounds when vibrated by air.

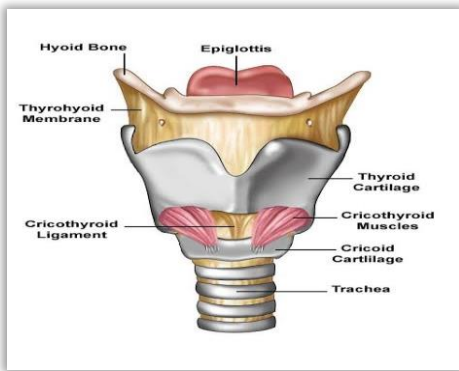


Figure 1.2. Larynx

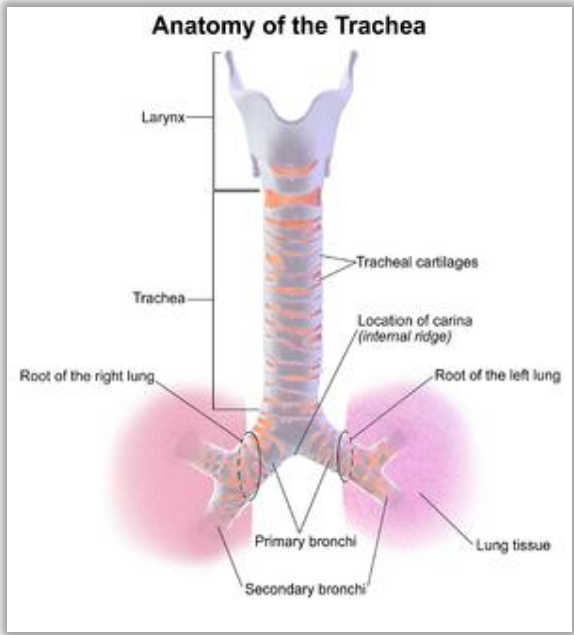


Figure 1.3. Anatomy of Trachea

TRACHEA is the narrow and cylindrical tube about five inches long located right after the larynx. The trachea functions as the passageway for air going in and out of the lungs.

BRONCHI are two air tubes that branch off of the trachea and carry air directly into the lungs.

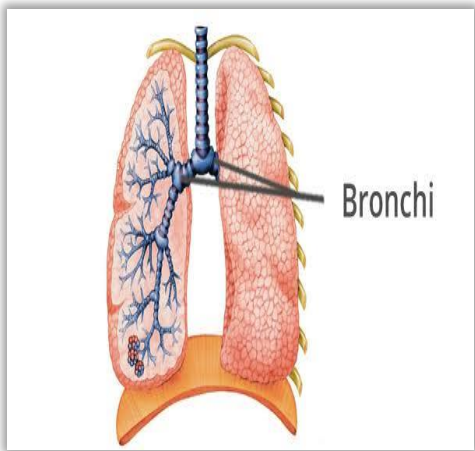


Figure 1.4. Bronchi

BRONCHIOLES are the smallest subdivisions that are less than 1 mm in diameter and which lack cartilage tissue. There are about 30,000 bronchioles in each lung. At the distal end of the bronchiole are the air sacs.

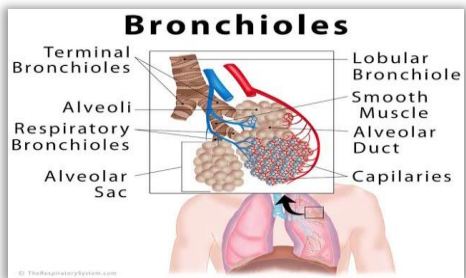


Figure 1.5. Bronchioles

AIR SACS look like bunches of grapes. The air sac is considered the functioning system of the lungs. Each air sac is composed of several microscopic out pocketing called alveoli (Sing. alveolus).

ALVEOLI permit the exchange of gases, oxygen and carbon dioxide through their thin walls. The alveoli are surrounded by a network of capillaries where the inspired gases pass through. Average adult lungs have approximately 3 million alveoli.

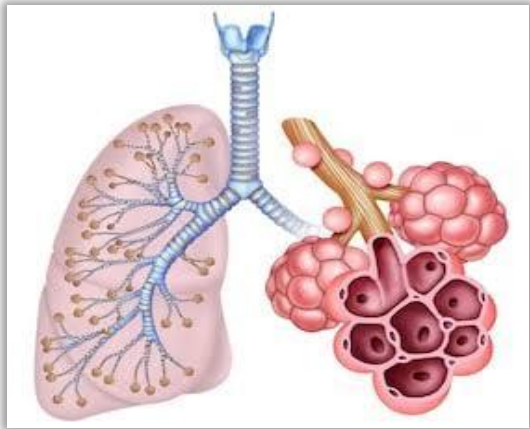


Figure 1.6. Air Sacs



Figure 1.7. Lungs

LUNGS are the cone-shaped organs of respiration in humans. Humans have two lungs that completely fill the spaces of the chest cavity and are surrounded by the rib cage.

DIAPHRAGM is a sheet of smooth muscle located below the lungs. It works with the lungs during the breathing process.

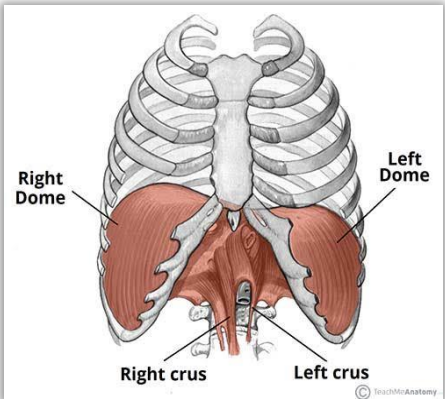


Figure 1.8. Diaphragm

THE BREATHING PROCESS

During breathing, the diaphragm, abdominal muscles, and intercostal muscles play important roles. The intercostal muscles are the muscles that are connected to the ribs. The breathing process has two phases: inhalation phase and exhalation phase.

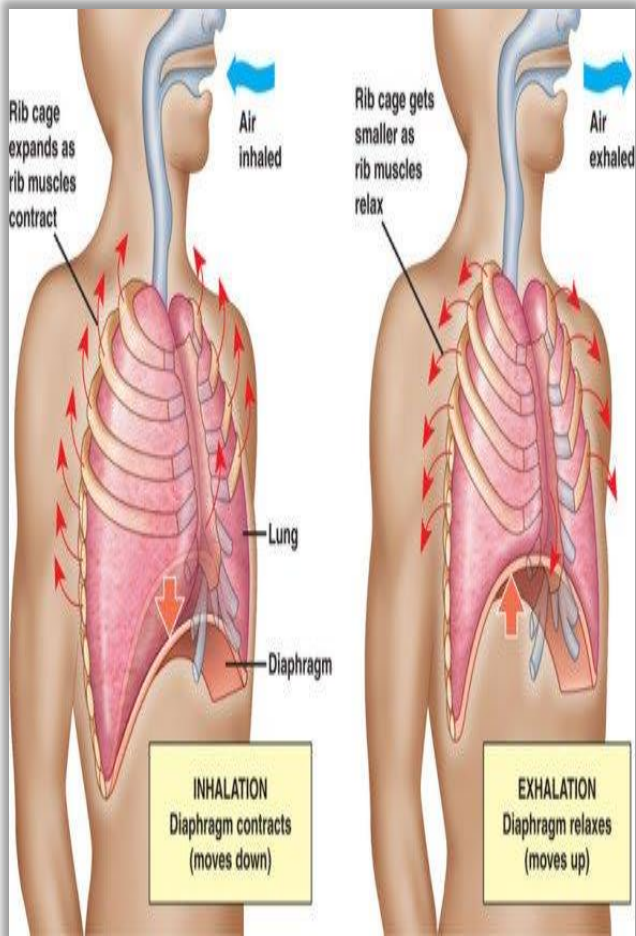


Figure 1.9. The diagrammatic illustration of the breathing process

INHALATION PHASE

During the inhalation phase the diaphragm contracts. A contracted diaphragm is flat in position. Then the intercostal muscles contract. The contraction causes the rib cage to move upward and outward. The volume of the thoracic cavity becomes larger than it was before. This is followed by the expansion of the lungs. This makes the air pressure within the alveoli (called intrapulmonic pressure) lesser than atmospheric pressure (air pressure outside the lungs). Consequently, the air naturally flows from outside the body into the respiratory passages and into the alveoli.

EXHALATION PHASE

During the exhalation phase, the diaphragm relaxes; that is, resuming its dome shape position. The intercostal muscles likewise relax and the ribcage moves down and in. Thus, the volume of the thoracic cavity decreases, the lungs recoil or deflate. This makes the air pressure within the alveoli (called intrapulmonic pressure) to increase above atmospheric pressure. Consequently, air

naturally flows from the inside of the body to outside.

PNEUMONIA is an infection of the alveoli caused by many kinds of both bacteria and viruses. Tissue fluids accumulate in the alveoli reducing the surface area exposed to air. If enough alveoli are affected, the patient may need supplemental oxygen.

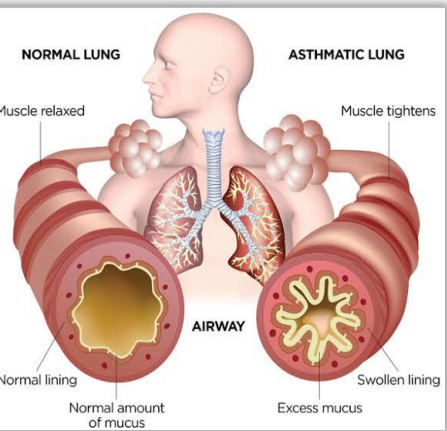


Figure 2.1. Asthmatic Lung
The gradual loss of gas-exchange areas forces the heart to work harder in order to satisfy the body's needs for oxygen. The added strain can lead to heart failure.



Figure 2.3. Thrombosis

ATHEROSCLEROSIS is a condition in which the arterial wall thickens as the result of a build-up of fatty materials such as cholesterol - a low-density lipoprotein. It is commonly referred to as the hardening or furring of the arteries.

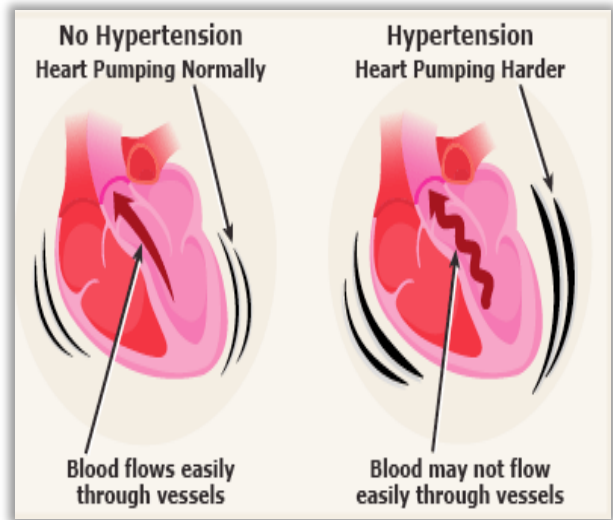


Figure 2.5. Hypertension

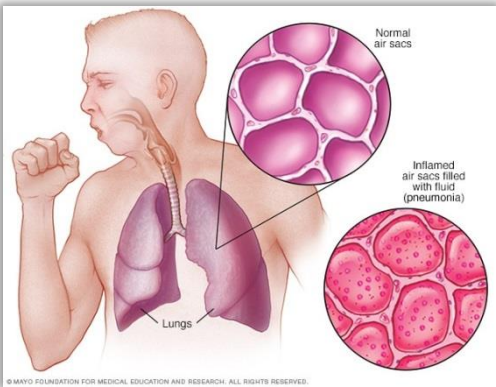


Figure 2.0. Pneumonia
ASTHMA is the periodic constriction the bronchi and bronchioles that makes breathing difficult. Airborne irritants to which the person is allergic to can trigger

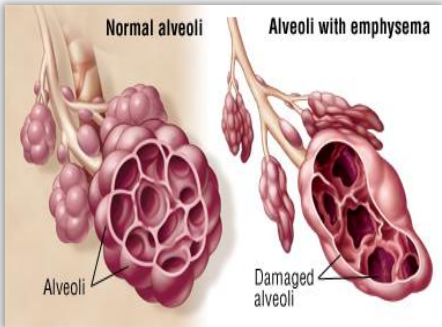


Figure 2.2. Alveoli with emphysema

THROMBOSIS is a condition where the blood in an unbroken blood vessel clots easily. This can lead to strokes or heart attacks.

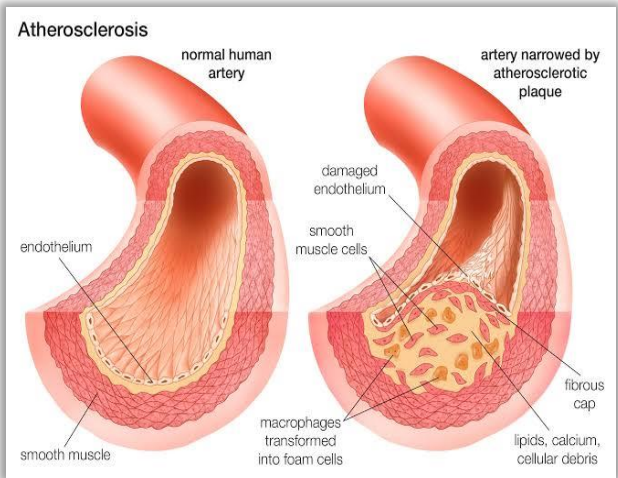


Figure 2.4. Atherosclerosis

HYPERTENSION more commonly known as high blood pressure, occurs when blood pressure remains consistently high. Hypertension is the major cause of stroke but can also cause damage to the kidneys, coronary arteries, and eyes.

NAME: _____ YEAR AND SECTION : _____

ACTIVITY 1: WHAT A BUNCH OF GRAPES!

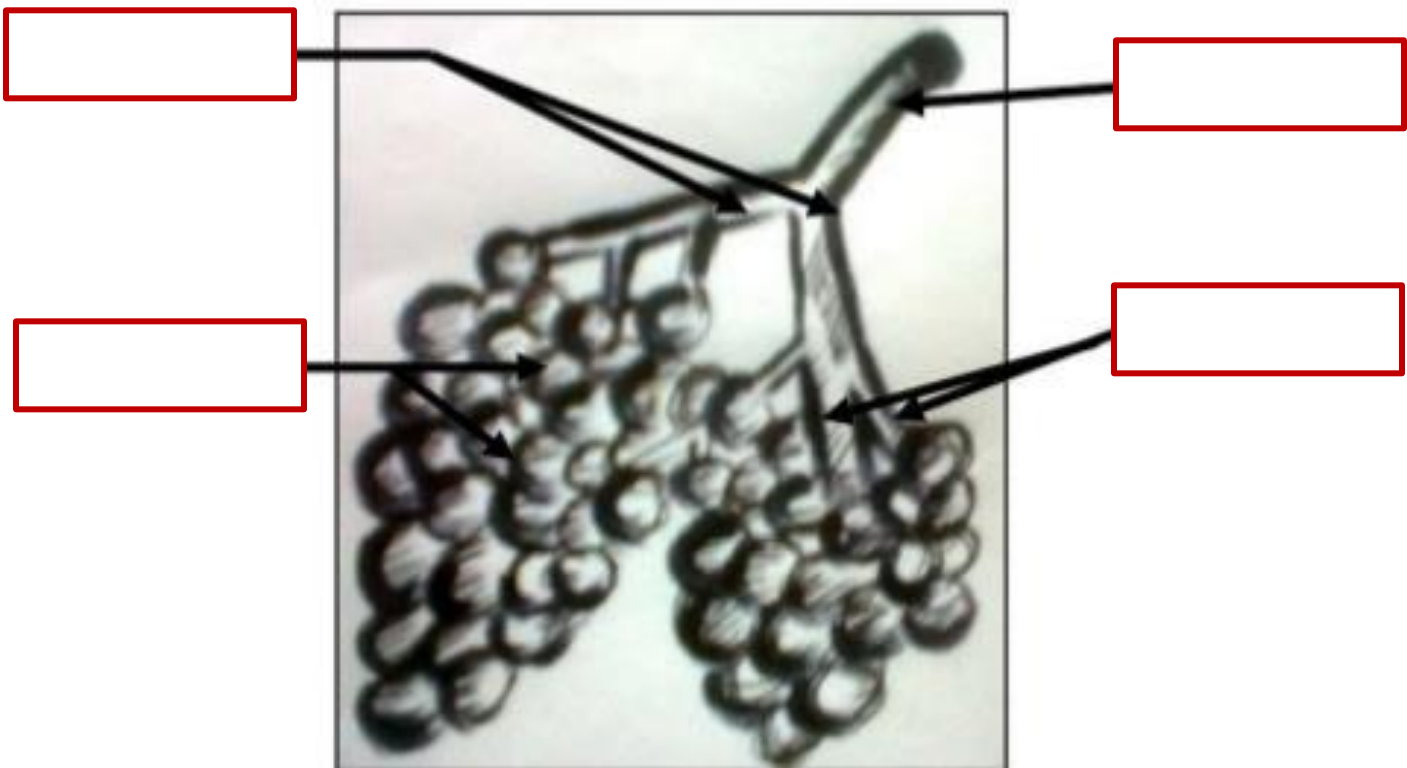
PART A

OBJECTIVE:

- 1. Identify the key parts of the breathing system

PROCEDURES:

- 1. Hold up the bunch of grapes. Let the bunch of grapes represent the breathing system.
- 2. Locate the parts of the breathing system: the main stem as the trachea, the large branching stems as the bronchi, and all the little stems as the bronchioles. The individual grapes are the airsacs or alveoli.
- 3. One by one, gently take out some of the grapes to expose more of the branching stems (bronchioles). Observe its structure.
- 4. Trace the pathway of oxygen using the “Bunch of Grapes” model. Note that air moves from the nose (nasal cavity) and mouth (oral cavity) to the trachea, bronchi, bronchioles, and then into the alveoli (air sacs). The air we breathe carries the gas oxygen. When we breathe, the oxygen goes to the lungs.



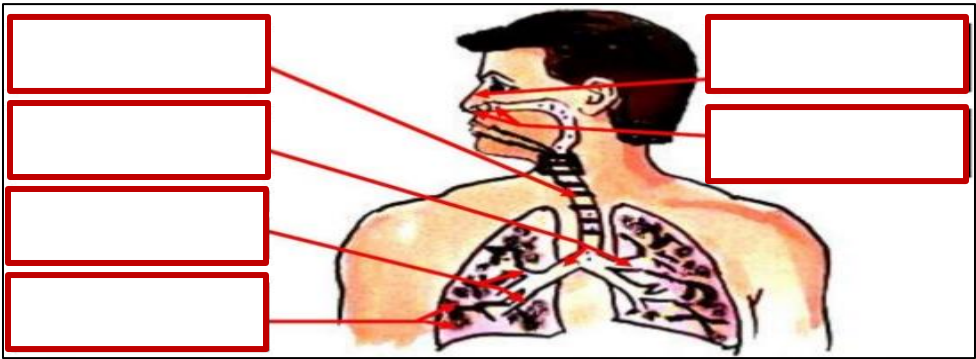
PART B

OBJECTIVE:

- 1. Describe the function of each part of the breathing system

PROCEDURE:

- 1. Refer to the diagram, and check your understanding of the breathing system by labeling each part and giving its functions in the box corresponding to the part.



GUIDE QUESTIONS:

- 1. What does each part of the “Bunch of Grapes” model represent, in relation to the breathing system?

- 2. How will you describe the pathway of oxygen in the breathing system?

- 3. What will happen if one part of the system fails to carry out its function properly?

ACTIVITY 2:

BOTTLED BALLOONS

OBJECTIVES:

- 1.1. Explain how the lungs work
- 1.2. Describe how the movement of the diaphragm helps the air go in and out of the lungs

MATERIALS:

1 two-liter empty plastic bottle 3 balloons (1 big, 2 small)
1 sturdy straw 1 pair of scissors 5 rubber bands

PROCEDURE:

1. Using a pair of scissors, cut the bottom out of the 2-liter plastic bottle.
2. Create two holes that are apart from each other in the cap of the plastic bottle. Make sure that each hole is just big enough for a straw to fit through.
3. Stick the two straws through the two holes of the bottle cap.
4. Place one balloon on the end of each straw, and secure them with rubber bands, as shown in the figure below.
5. Stick the balloon ends of the straws through the bottle opening and screw the lid on tightly.
6. Stretch out the larger balloon and place it over the open bottom of the bottle. Secure it with the rubber band as tightly as possible. Refer to the diagram of the finished lung model below.
7. Pull the larger balloon down; that is, away from the bottle, in order to blow up the two small balloons.
8. Push the larger balloon towards the bottle in order to let the air out of the two small balloons.
9. Write down your observations.

GUIDE QUESTIONS:

Q1. What does each part of the constructed lung model represent?

Q2. What happens as you pull down the balloon at the bottom of the model?

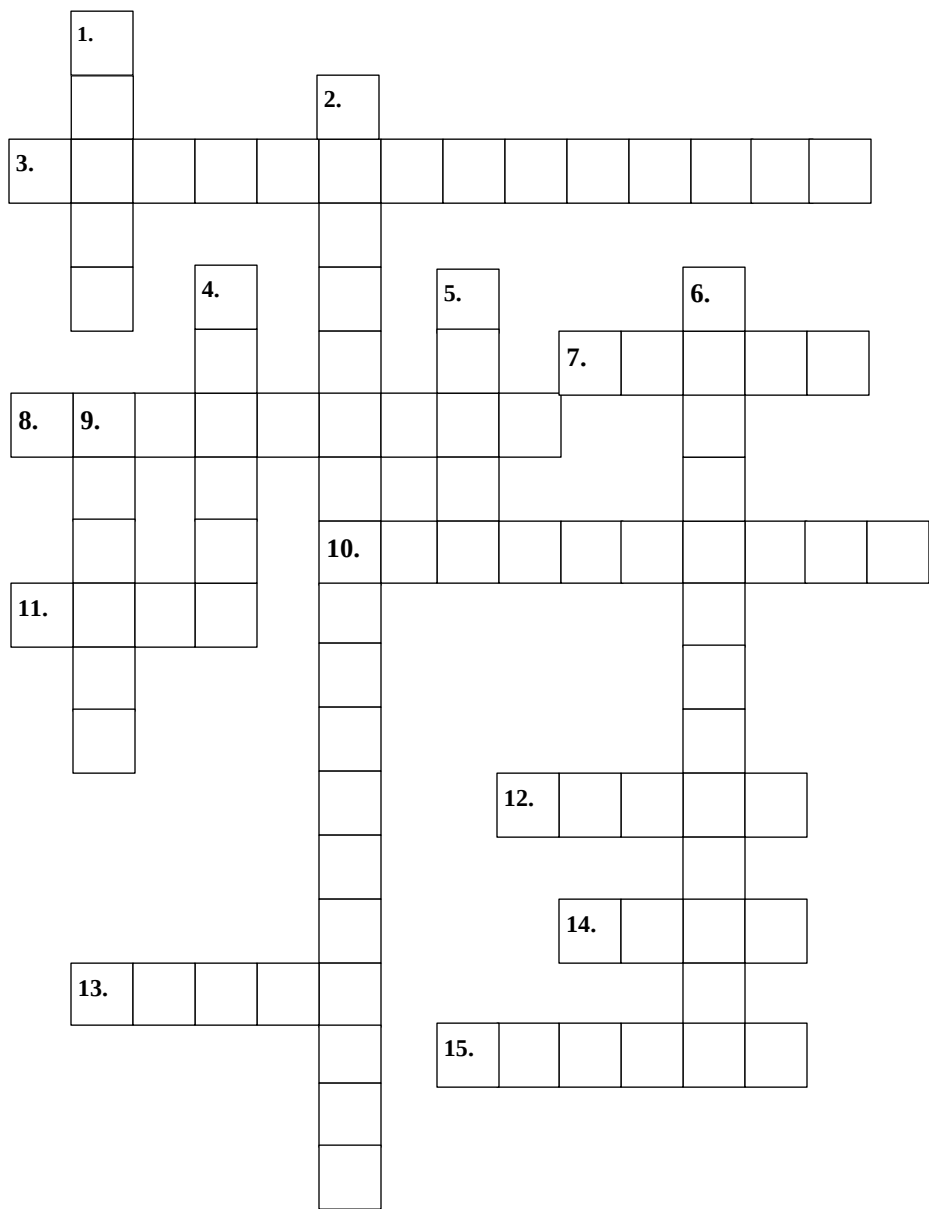
Q3. What happens as you push up the balloon?

Q4. How does the movement of the diaphragm cause the air to go in and out of the lungs?

ACTIVITY 3:

CROSSWORD PUZZLE

Directions: Solve the crossword puzzle. Fill in each box the letter of the correct answers.

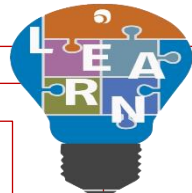


Across

- 3- The place where the nutrients enter the blood.
- 7- The pump in the circulatory system.
- 8- A small vessel that connects veins and arteries.
- 10- The circulatory system _____ nutrients, gases, liquids, and heat around the body.
- 11- The circulatory system transports this, which helps regulate temperature.
- 12- The liquid in the circulatory system.
- 13- The place where oxygen enters the blood and carbon dioxide leaves the blood.
- 14. A vessel that moves blood to the heart.
- 15- A gas that is transported in arteries from the lungs to the rest of the body.

Down

- 1- The heart _____ blood.
- 2- The heart, blood and vessels.
- 4- Blood in arteries is _____ red because it is rich in oxygen.
- 5- The largest artery in the body.
- 6- A waste gas that is transported in veins from the body to the lungs via heart.
- 9- A vessel that moves blood away from the heart.



Concept Micro

Respiratory system provides passageway of air in our body and it sustains life to each living organism, therefore we must take care of it to maintain its optimum health.

HOMEWORK

1. Research the following terminologies:
 - 1.1. Chromosomes
 - 1.2. Monohybrid
 - 1.3. Dihybrid
 - 1.4. Punnett square
 - 1.5. Haploid and Diploid
2. Have an advance study about genetic reasoning.
3. Prepare pictures of your parents or relatives.



TEST YOURSELF!

TEST I. MULTIPLE CHOICE. Write the letter of the best answer on the blank before each number.

- _____ 1. What happen when you breathe in?
 - a. Your diaphragm expands and your rib cage contracts.
 - b. Nothing happens.
 - c. Your diaphragm explodes and your rib cage contracts.
 - d. Your diaphragm contracts and your rib cage expands.
- _____ 2. Which of these doesn't occur when air flows into the lungs?
 - a. the air is humidified
 - b. the air is cooled
 - c. the air is filtered
 - d. the air disappeared
- _____ 3. It is the organ that connects the pharynx and trachea.
 - a. Nose
 - b. Throat
 - c. Pharynx
 - d. Larynx
- _____ 4. What is the function of the respiratory system?
 - a. the body's breathing system.
 - b. the body's system of nerves.
 - c. the body's food processing system.
 - d. the body's blood transporting system.
- _____ 5. Which of the following does not belong to the group?
 - a. asthma
 - b. hypertension
 - c. pneumonia
 - d. emphysema



DEFINITION OF TERMS

GLOSSARY OF TERMS

ALVEOLI permit the exchange of gases, oxygen and carbon dioxide through their thin walls.

ASTHMA is the periodic constriction of the bronchi and bronchioles that makes breathing difficult.

BRONCHI are two air tubes that branch off of the trachea and carry air directly into the lungs.

DIAPHRAGM is a sheet of smooth muscle located below the lungs.

EMPHYSEMA is a disorder in which the delicate walls of the alveoli break down, reducing the gas- exchange area of the lungs.

LARYNX is the organ that connects the pharynx and the trachea.

LUNGS are the cone-shaped organs of respiration in humans

NOSE is the gateway of the respiratory system.

PHARYNX is the narrow space located right after the mouth and nose.

RESPIRATORY SYSTEM is the biological system of an organism that introduces respiratory gases to the interior and performs gas exchange.



REFERENCES

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ELECTRONIC SOURCE

<http://depedk12manuals.blogspot.com/2016/06/grade-9-learners-module-and-teachers.html?=1>